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Chapter 20

GELFAND-P000981

14/IX-81. Gelfand Sem.

I.M.G. On contragredient Lie algebras

21/IX-81. Gelfand Sem.

S. Gelfand.

Reading on Lie algebras:

1. Dixmier or Serre, part 3 (do not read 1 and 2 under any circumstances)

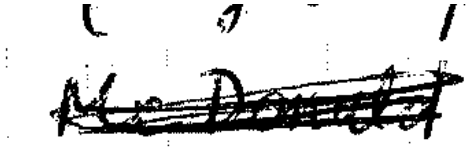
[Below the name Dixmier, linked by a short vertical line:] Enveloping algebras, ch. 1 — textbook literature.

2. Humphrey, Lect. Notes, 831

highest-weight modules for semisimple Lie algebras.

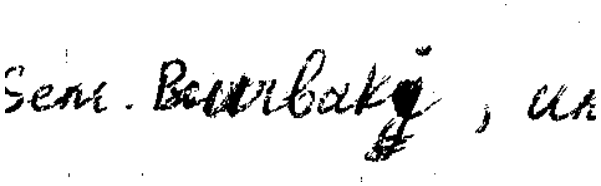
(Verma modules)

3. [Crossed out: Mc Donald[?]] Macdonald Sem. Bourbaki[?], June 1981.



Uncertain reading. Mc Donald

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Uncertain reading. Bourbaki

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4. Works of Kac and Gabber

5. I.M. Gelfand et al. Funct. Anal., 5:1; 10:2

(Bernstein, [crossed out: И[?]], I.M.G., S.I.G.)



Uncertain reading. И

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6. Jantzen, Lect. Notes, 750

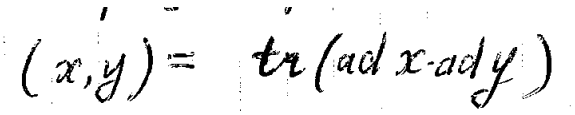
7. Kac, Kazhdan Advances Math., 1979, 34, $N = 1$.

I.M.G. h — vector space with basis $h_1, \ldots, h_n$

g — semisimple Lie algebra

$$\begin{aligned} \operatorname{ad} x \colon g &\longrightarrow g \\ y &\longmapsto [x, y] \end{aligned}$$

Cartan scalar product: $(x, y) = \operatorname{tr}(\operatorname{ad} x - \operatorname{ad} y)$?



Uncertain reading. $(x, y) = \operatorname{tr}(\operatorname{ad} x - \operatorname{ad} y)$

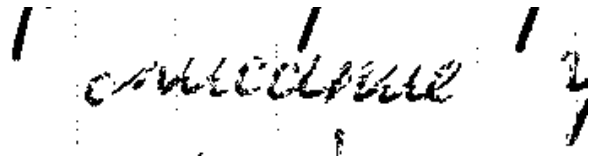
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(Check on the matrix algebra)

[Added below the line:] and the orthog. alg. of order 3.

I.M.G. Infinite-dimensional representations lie close to the standard representation of the Lorentz group.

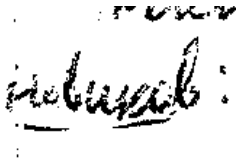
Idea: into physics, since the shift? of the Lorentz group is approximate (at our scales this is negligible)



Uncertain reading. shift

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Novikov?: a small shift should give the neutrino mass

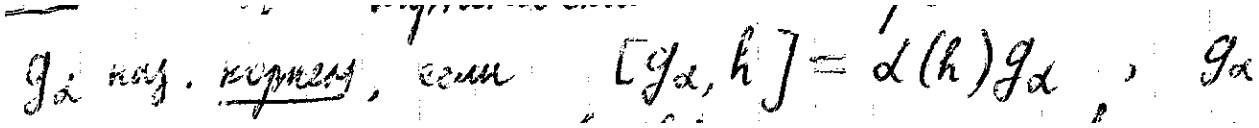


Uncertain reading. Novikov

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I.M.G. h — Cartan subalgebra

g_α is called a root if $[g_\alpha, h] = \alpha(h) g_\alpha$, g_α?



Original source details: GELFAND-P000981

Uncertain reading. g_α is called a root if $[g_\alpha, h] = \alpha(h) g_\alpha$, g_α

($\alpha(\cdot)$ — linear function on h, called a root)

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[source number: 12]

On h a scalar product appears (dual to the Cartan one).

Then we obtain $(\alpha, \alpha) > 0$.

There exist roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that if α is another root, then

$$\alpha = \sum n_i \alpha_i, \qquad n_i \in \mathbb{Z}.$$

Here all the n_i have the same sign!

If all $n_i \geq 0$, the root is pos.

— ” — $n_i \leq 0$ — ” — neg.

Necessarily [crossed out: that?] all the n_i have the same sign!!



Uncertain reading. that

Original source details: GELFAND-P000982

They can be ordered

$\exists$ a greatest root α_0

Coroot:

$$\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}$$

The set of coroots is again a root system (another algebra — Langlands dual).

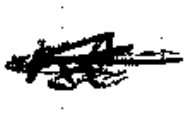
We construct the reflection $w_\alpha \colon h \longrightarrow h$, $w_\alpha(\lambda) =$ [crossed out: unresolved sign] $\lambda - (\alpha^\vee, \lambda)\alpha$



Uncertain reading. unresolved sign

Original source details: GELFAND-P000982

We identify h and h^* [crossed out: unresolved word] by means of the metric



Uncertain reading. unresolved word

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[Above the following line, crossed out: α?]



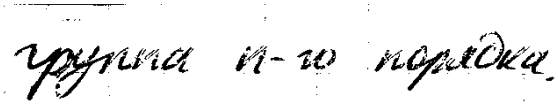
Uncertain reading. α

Original source details: GELFAND-P000982

w_α — reflection in the plane perpendicular to α.

$$w_\alpha(\alpha) = -\alpha.$$

The group generated by these reflections is finite. It is called the Weyl group W.
For $sl(n)$ this is the [crossed out: gr[?]] symmetric [crossed out: of permutations[?]] group of order n.



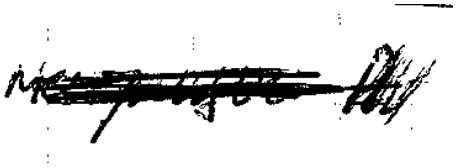
Original source details: GELFAND-P000982

Diagram and typed description. After "group" the manuscript reads "of order n".



Original source details: GELFAND-P000982

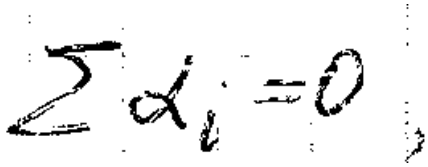
Uncertain reading. gr



Original source details: GELFAND-P000982

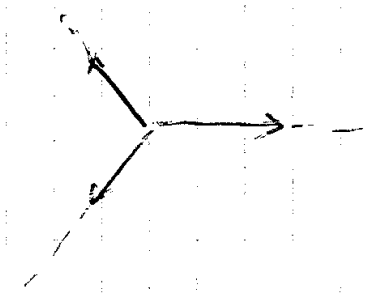
Uncertain reading. of permutations

$sl(3) \Rightarrow h$ — a 2-dimensional space
$\sum \alpha_i = 0$, we draw them all in space, but they lie in the plane $\sum_{i=1}^{3} x_i = 0$



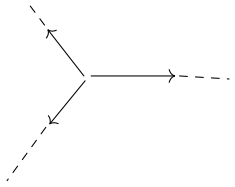
Original source details: GELFAND-P000982

Diagram and typed description. The first sum has no written limits.



Original source details: GELFAND-P000982

Diagram and typed description. ← we obtain 3 vectors at an angle of $120°$.



Reduce [crossed out: unresolved sign] to reflections: permutations — to transpositions.



Original source details: GELFAND-P000982

Uncertain reading. unresolved sign

reflections of adjacent ones — simple roots.

All roots are obtained from the simple ones by transformations of the group W.

$a_{ij} = (\alpha_i^\vee, \alpha_j)$ — the Cartan matrix.

Claim. $a_{ij} \in \mathbb{Z}$; $a_{ij} \leq 0$; $a_{ij} = 0 \Rightarrow a_{ji} = 0$

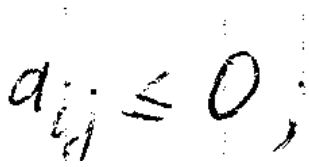


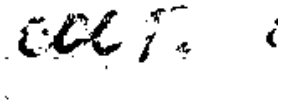
Diagram and typed description. The inequality $a_{ij} \leq 0$ has no written restriction on the indices.

Original source details: GELFAND-P000982

(a_{ij}) is positive definite.

Conversely: to every such matrix corresp.? a Lie alg.

Uncertain reading. corresp.



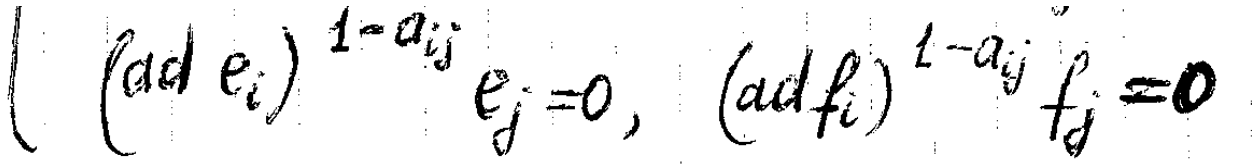
Original source details: GELFAND-P000982

Put $\alpha_i^\vee = h_i$.

Then we introduce the Lie alg. defined by the following relations:

GELFAND-P000983

$$\begin{cases} [e_i, f_j] = \delta_{ij} h_i \\ [h_i, h_j] = 0 \\ [h_i, e_j] = a_{ij} e_j, \quad [h_i, f_j] = -a_{ij} f_j \\ (\operatorname{ad} e_i)^{1-a_{ij}} e_j = 0, \quad (\operatorname{ad} f_i)^{1-a_{ij}} f_j = 0 \end{cases} \tag{*}$$



Original source details: GELFAND-P000983

Diagram and typed description. The last line of the system has no written condition on the indices.

V — a g-module

$\lambda \in h^*$ is called a weight if it is an eigenvalue, i.e. $\exists v \in V \setminus \{0\}$

$$hv = \lambda(h)v.$$

V_λ — the subspace of eigenvectors with weight λ.

Example of a g-module: g with ad.

$$\dim V < \infty \quad \Rightarrow \quad V = \bigoplus V_\lambda$$

and all λ lie in the lattice generated by the roots (coroots)

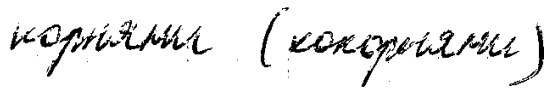


Diagram and typed description. The manuscript reads “roots (coroots)”.

Original source details: GELFAND-P000983

Ordering λ: lexicographically by the coordinates generated by the simple roots.

$\exists!$ finite-dimensional rep. generated by a highest-weight vector.

[Crossed out: $\forall^?$] α — a root, λ — a weight, v_λ has weight $\lambda \Rightarrow$

$e_\alpha v_\alpha$ has weight $\alpha + \lambda$.

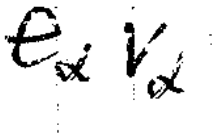


Diagram and typed description. The manuscript writes v_α after e_α.

Original source details: GELFAND-P000983

Uncertain reading. $\forall$



Original source details: GELFAND-P000983

Proof.

$$\begin{aligned} h(e_\alpha v_\alpha) &= e_\alpha(hv_\alpha) + [h, e_\alpha]v_\alpha \\ &= \lambda e_\alpha v_\alpha + \alpha e_\alpha v_\alpha = \\ &= (\lambda + \alpha) e_\alpha v_\alpha \end{aligned}$$

Cartan’s thm.

[Crossed out: $\exists!^?$; added above the line inside an outline: $\forall\lambda \geq 0\ \exists!$] finite-dimensional rep.
[crossed out: if [remainder unresolved]] such that

$e_\alpha V_\lambda = 0, \forall \alpha \in \Delta_+$ (pos. roots)

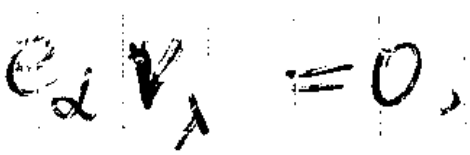
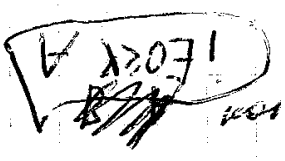


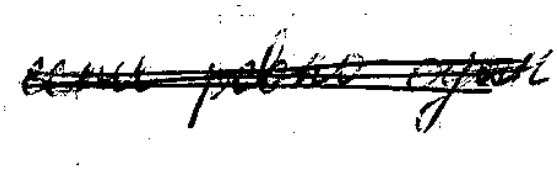
Diagram and typed description. The letter V in V_λ is retraced.

Original source details: GELFAND-P000983



Uncertain reading. ∃!

Original source details: GELFAND-P000983



Uncertain reading. if [remainder unresolved]

Original source details: GELFAND-P000983

(or $\sim \forall$ simple).

Introduce the group e^{λ} as follows:

$$e^{(\lambda+\mu)} = e^{\lambda} e^{\mu}$$

e — a letter.

$$e^{\sum n_i \alpha_i} = (e^{\alpha_1})^{n_1} \cdots (e^{\alpha_n})^{n_n}$$

Character of the representation:

$$\operatorname{ch} V \underset{\text{def}}{=} \sum_{\mu} (\dim V_{\mu}) e^{\mu}$$

Introduce [crossed out: root [fragment]?] the following weight ρ:

$$\rho(h_i) = 1, \qquad i = 1, \ldots, n$$



Uncertain reading. root [fragment]

Original source details: GELFAND-P000983

$$h_i = \alpha_i^{\vee}$$

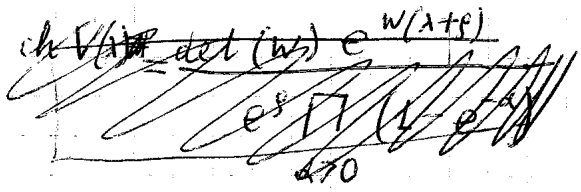
Weyl formula: for the rep. with highest weight λ:

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[source number: 13]

[Crossed-out formula:]

$$\operatorname{ch} V(\lambda) = \frac{\det(w) e^{w(\lambda+\rho)}}{e^{\rho} \prod_{\alpha>0} (1 - e^{-\alpha})}.$$



Uncertain reading. In the cancelled formula no sum is visible before $\det(w)$. The strokes after $V(\lambda)$ overlap.

Original source details: GELFAND-P000984

$$\operatorname{ch} V(\lambda) = \frac{\sum_{w\in W} \det(w) e^{w(\lambda+\rho)}}{e^{\rho}\prod_{\alpha>0}(1-e^{-\alpha})}.$$

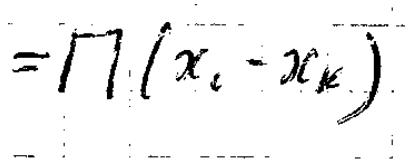
$w(\lambda+\rho)$ — result of the action of w on the weight $\lambda+\rho$.

Hence, in particular,

$$\sum \det(w) e^{w\rho} = e^{\rho}\prod_{\alpha>0}(1-e^{-\alpha}).$$

For matrices this is the formula

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix} = \prod (x_i - x_k).$$



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Diagram and typed description. The product has no written index limits.

Rudakov. Kac–Moody algebras

A — an $n \times n$ matr.

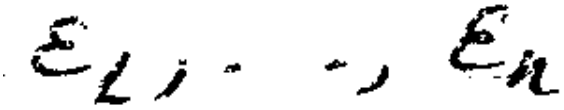
and a set of generators

$$e_1, \ldots, e_n, \quad f_1, \ldots, f_n, \quad h_1, \ldots, h_n, \quad \text{[crossed out] } \varepsilon_1, \ldots, \varepsilon_n$$



Original source details: GELFAND-P000984

Uncertain reading. The short entry before ε_1 is crossed out. Its reading remains unresolved.



Original source details: GELFAND-P000984

Diagram and typed description. The subscript on the first visible ε reads 1.

and rels.

$$\begin{cases} [h_i, h_j] = 0 \\ [h_i, e_j] = a_{ij} e_j \\ [h_i, f_j] = -a_{ij} f_j \\ [e_i, f_j] = \delta_{ij} h_i \end{cases}$$

$$\begin{cases} [\varepsilon_i, \varepsilon_j] = 0 \\ [\varepsilon_i, e_j] = \delta_{ij} e_j \\ [\varepsilon_i, f_j] = -\delta_{ij} f_j \\ [\varepsilon_i, h_j] = 0 \end{cases}$$

$F(A)$ — Lie algebra generated by these generators and relations.

$$F(A) \supset \widetilde{H} = \langle h_i, \varepsilon_i \rangle.$$

I — max ideal in $F(A)$ such that the image of $\{e_i, f_i, h_i, \varepsilon_i\}$ in $F(A)/I$ is a linearly indep. system of vectors.

It exists and is uniquely determined (max "non-injuring").

$$\widetilde{G}(A) \stackrel{\text{def}}{=} F(A)/I.$$

GELFAND-P000985

(2) — matrix of order 2 with trace 0.

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

[Crossed out: short entry unresolved.]



Uncertain reading. short entry unresolved

Original source details: GELFAND-P000985

Add. max non-injuring system

Criterion for finite dimension

matr. is symmetrizable (made symmetric by mult. by a diagonal one), and positive definite, 2 on the main diagonal

$$\widetilde{G}(A) \supset \underset{\text{center}}{Z}$$

$$Z \subset \langle h_i, \varepsilon_i \rangle$$

We still have to quotient by a max subspace in Z that does not intersect the linear span of $\{h_i\}$.

$$J \subset Z$$

(a_{ij}) is nonsingular $\Rightarrow$ this is no longer needed

(a_{ij}) is singular $\Rightarrow$ the roots are linearly dependent. We add [crossed out: some[?] remainder unresolved], to make them linearly independent.

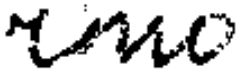
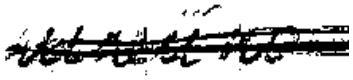


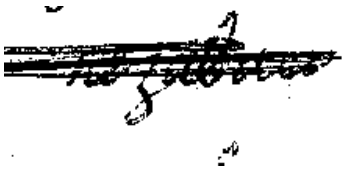
Diagram and typed description. After the cancellation the manuscript reads «что».

Original source details: GELFAND-P000985



Uncertain reading. some

Original source details: GELFAND-P000985



Original source details: GELFAND-P000985

Uncertain reading. remainder unresolved

For this we formally enlarge h with room to spare

$$J \subset Z, \qquad J \cap \langle h_i \rangle = 0.$$

$$G(A) \overset{\text{def.}}{=} \widetilde{G}(A)/J_{\max}.$$

$$H = \langle h_i, \varepsilon_i \rangle.$$

Define the set of vectors $x \in G(A)$ that are eigenvectors.

$$G_\alpha = \{x \in G(A),\ \forall h \text{ [crossed out]} \in H,\ [h, x] = \alpha(h)x\}.$$



Original source details: GELFAND-P000985

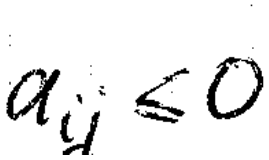
Uncertain reading. A symbol between h and $\in$ is crossed out. Its reading is unresolved.

$$\alpha \in H^*$$

Roots: those α for which G_α are nonzero.

A is a Cartan matrix if

1) $a_{ij} \in \mathbb{Z}, \qquad a_{ii} = 2, \qquad a_{ij} \leq 0$



Original source details: GELFAND-P000985

Diagram and typed description. There is no written index condition on $a_{ij} \leq 0$.

2) $a_{ij} = 0 \Rightarrow a_{ji} = 0$

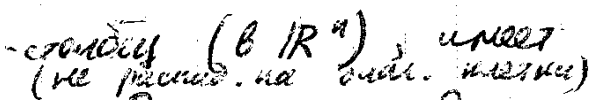
[Crossed out before the addition: entry unresolved.]



Original source details: GELFAND-P000985

Uncertain reading. entry unresolved

[Added above the lemma: Let A be irreducible (does not split into diag. blocks[?]).]



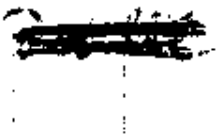
Original source details: GELFAND-P000985

Uncertain reading. does not split into diag. blocks

[Written above: u is a column vector (in $\mathbb{R}^n$), so $Au \in \mathbb{R}^n$ makes sense.]

Lemma.

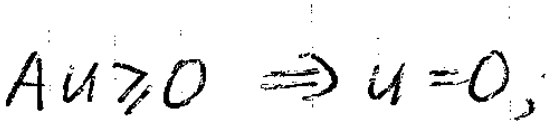
(I) If $Au \geq 0$, then $u > 0$ or $u = 0$; [crossed out: entry unresolved] $\det A \neq 0$.



Original source details: GELFAND-P000985

Uncertain reading. entry unresolved

(II) $Au \geq 0 \Rightarrow u = 0; \quad \exists u > 0, Au = 0; \quad \operatorname{rank} A = n - 1$



Original source details: GELFAND-P000985

Diagram and typed description. To the right of the arrow the manuscript reads $u = 0$.

$$\text{(III)} \quad Au \geq 0,\ u \geq 0 \ \Rightarrow\ u = 0; \quad \exists u > 0 :\ Au < 0$$

Assertion: exactly one of (I)–(III) holds.

GELFAND-P000986

[source number: 14]

(I) — finite-dimensional Lie algs.; they are classified

(II) — extended.

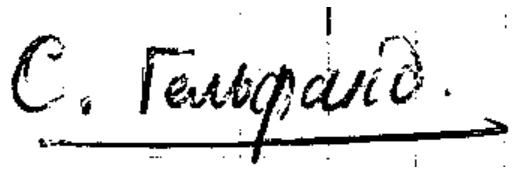
(III) — all the others.

I. M. G. Ramanujan's identities:

$$\frac{1}{\prod_{n=1}^{\infty}(1-q^{5n-2})(1-q^{5n-3})} = \sum_{m=0}^{\infty} \frac{q^{m(m+1)}}{(1-q)\cdots(1-q^m)}$$

$$\frac{1}{\prod_{n=1}^{\infty}(1-q^{5n-1})(1-q^{5n-4})} = \sum_{m=0}^{\infty} \frac{q^{m^2}}{(1-q)\cdots(1-q^m)}$$

S. Gelfand.



Original source details: GELFAND-P000986

Diagram and typed description. The line under the name ends in a right-pointing arrow.

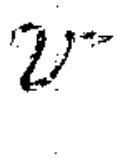
Verma module.

(see Dixmier, Jantzen)

In $G(A)$ there are generators $e_i, f_i, h_i \qquad H = \{h_i\}$

$\forall \lambda \in H^*$ we construct a module generated by one vector $v^{-?}$ with the conditions

$$(*) \quad \begin{cases} e_i v_0 = 0 \\ h v_0 = \lambda(h) v_0 \end{cases}$$



Uncertain reading. v^-

Original source details: GELFAND-P000986

$$N_+ = \{e_i\}, \qquad N_- = \{f_j\}$$
$$N_+ v_0 = 0$$

$f_{i_1} f_{i_2} \cdots f_{i_k} v_0$ — we write out such letters.

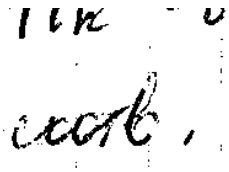
Impose rels. $(*)$ and also those dictated by f_j.

One readily obtains

$$h f_{i_1} \cdots f_{i_k} v_0 = [\lambda(h) - \alpha_{i_1}(h) - \cdots - \alpha_{i_k}(h)] f_{i_1} \cdots f_{i_k} v_0,$$

i.e. $f_{i_1} \cdots f_{i_k} v_0$ is also a weight vector

The module corresp.[?] to the char. λ is called $M(\lambda)$.



Uncertain reading. corresp.

Original source details: GELFAND-P000986

$$\mathrm{Hom}(M(\lambda), M(\mu)) = \begin{cases} 0 \text{ or} \\ \mathbb{C} \end{cases}$$

— and then every nontriv. homom. is an embedding

Theorem. If $\mathrm{Hom}(M(\lambda), M(\mu)) \neq 0$, then

$$1) \quad \lambda = \mu - \sum n_i \alpha_i \quad \text{for some integers } n_i \geq 0$$

$$2) \quad \lambda - \rho = w(\mu - \rho) \quad \text{for some } w \in W$$

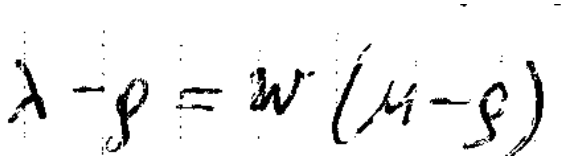


Diagram and typed description. Both occurrences of ρ are preceded by minus signs.

Original source details: GELFAND-P000986

(ρ is the half-sum of all pos. roots — see above).

$$3) \quad \lambda - \rho = s_{\delta_1} \cdots s_{\delta_k}(\mu - \rho)$$

(s_δ — reflections in any roots, not necess. simple)

and

$$- s_{\delta_i} \cdots s_{\delta_k}(\mu - \rho) + s_{\delta_{i-1}} s_{\delta_i} \cdots s_{\delta_k}(\mu - \rho) = m_i \delta_{i-1},$$
$$m_i \leq 0, \quad m_i \in \mathbb{Z}.$$

1)–3) are already sufficient for $\mathrm{Hom}(M(\lambda), M(\mu))$ to be nontrivial.

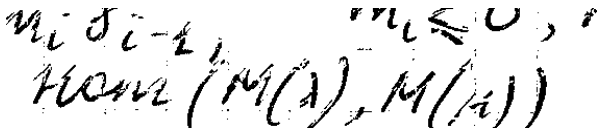


Diagram and typed description. In the final Hom the second weight reads μ.

Original source details: GELFAND-P000986

GELFAND-P000987

Problem (I.M.G.) (nontrivial)



Diagram and typed description. The first word of the heading reads “Problem”.

Original source details: GELFAND-P000987

(a_{ij}) — Cartan matr.; satisfies cond. 1) and 2)

What is the dimension of the root space?

For every root, lin. indep. words are constructed, etc.



Diagram and typed description. The abbreviation before “indep.” reads «лин.».

Original source details: GELFAND-P000987

28.9.81. Gelfand Sem.

Seifert (?) (Paris)



Diagram and typed description. The manuscript has “(?)” after “Seifert”.

Original source details: GELFAND-P000987



Diagram and typed description. A yellow comment icon appears to the right of the heading in the digital frame.

Original source details: GELFAND-P000987

(X, G) — an arbitrary transformation group

$\mathcal{P}(X)$ — the set of subsets of X

$\mathcal{F} \subset \mathcal{P}(X)$ is defined as follows:

[Crossed out above the start of the formula: entry unresolved.]



Uncertain reading. entry unresolved

Original source details: GELFAND-P000987

[Crossed out before S: entry unresolved.]



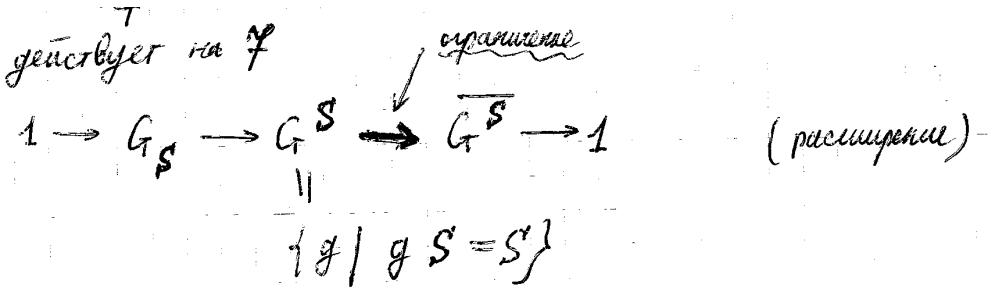
Uncertain reading. entry unresolved

Original source details: GELFAND-P000987

$$S \in \mathcal{F} \underset{\text{def}}{\iff} \forall x \notin S,\ \exists g \in G,\ gs = s,\ \forall s \in S,\ gx \neq x$$

G acts on $\mathcal{F}$

$$1 \longrightarrow G_S \longrightarrow \underset{\underset{\{g \mid gS=S\}}{\|}}{G^S} \xrightarrow{\text{restriction}} \overline{G^S} \longrightarrow 1 \qquad \text{(extension)}$$



Original source details: GELFAND-P000987

Diagram and typed description. The pointer from “restriction” marks the arrow from G^S to $\overline{G^S}$.

Here

$$G_S = \bigcap_s G_s, \qquad G_s = \mathrm{St}_G(s)$$

$\mathcal{F}$ has the following formal property:
it is closed under arbitrary intersections

$$S_i \in \mathcal{F},\ \forall i \ \Rightarrow \bigcap_{i \in I} S_i \in \mathcal{F}$$

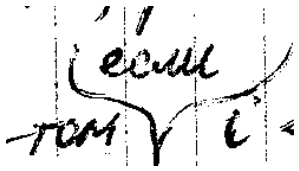
$(I, <)$ — a partially ordered set. Consider fns. $f : I \to \mathcal{F}$ with the property:

$$\{f : I \to \mathcal{F} \mid f(i) \subseteq f(j) \Longleftrightarrow i \leq j\}$$

Then G acts on this set.
Fixing $(I, <)$, we can in particular consider all such fns.

$$(\overline{\mathcal{F}}_I, G) = \{f : I \to \mathcal{F} : \text{ as above and moreover}$$

["If" is inserted above the line with an insertion mark.]



Original source details: GELFAND-P000987

Diagram and typed description. The word "if" is above the V-shaped mark.

$i < i'$ [crossed out: entry unresolved] a minimal chain (2 adjacent elem.) in $(I, <)$
$\Rightarrow f(i) \underset{\neq}{\subset} f(i')$ is also a min. chain $\}$



Original source details: GELFAND-P000987

Uncertain reading. entry unresolved



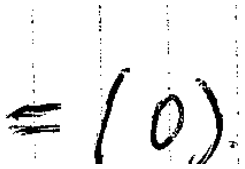
Original source details: GELFAND-P000987

Diagram and typed description. The inclusion sign has $\neq$ beneath it.

Then G also acts on the set of such fns.

From the formal property of stability under intersections

$$S \subseteq X \rightsquigarrow \underset{\text{(closure)}}{S^{\mathrm{cl}}} = \bigcap_{\substack{T \in \mathcal{F} \\ T \supseteq S}} T = (0).$$



Original source details: GELFAND-P000987

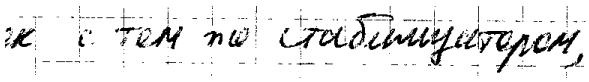
Diagram and typed description. The manuscript writes (0) to the right of the final equals sign.

GELFAND-P000988

[source number: 15]

$$x \in X \Rightarrow \text{define } \{x\}^{\mathrm{cl}} \underset{\text{def}}{=} \text{line passing through this point}$$

G [inserted: acts] transitively $\Rightarrow \{x\}^{\mathrm{cl}}$ is the set of points with the same stabilizer as x



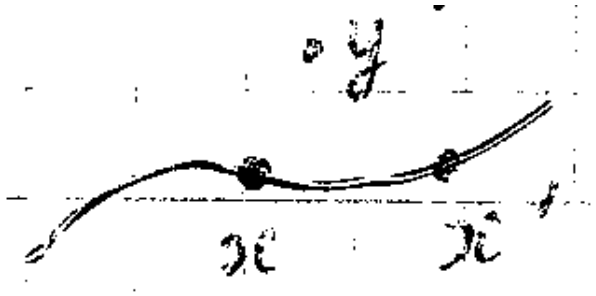
Original source details: GELFAND-P000988

Diagram and typed description. The manuscript reads “with the same stabilizer”.

G_x notation for the stabilizer.

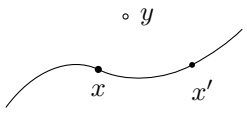
Idea: single transitivity $\Rightarrow$ multiple transitivity

$$\exists x : \ \{x\}^{\mathrm{cl}} \underset{\neq}{\supseteq} x \Rightarrow \text{the group cannot act doubly transitively}$$



Original source details: GELFAND-P000988

Diagram and typed description. Two marked points x, x' on the curve and a point y above it.



$(x, y) \mapsto (x, x')$ is impossible, since $\{x\}^{\mathrm{cl}}$ is pres.?



Original source details: GELFAND-P000988

Uncertain reading. pres.

On $\overline{\mathcal{F}}_I$ it can already act doubly transitively

Example

doubly trans. on $F \iff$ trans. on $(F \times F) \setminus \Delta$

$$\forall x \in F, \quad G_x \text{ must act transitively on } F \setminus \{x\}$$

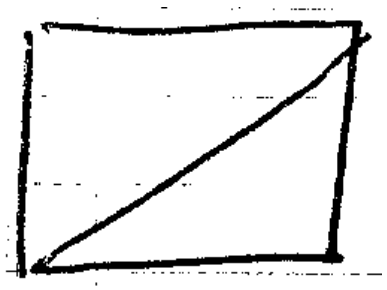
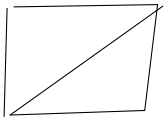


Diagram and typed description. A quadrilateral with a rising diagonal and a small gap in its upper left corner.



Original source details: GELFAND-P000988

The end will be as follows: for

B, N, T, W, P

an axiomatic generalization is given, for which the Bruhat decomp. makes sense.

Under certain conditions on the action:

(i)

$$\mathcal{F}_{\max} = \{\max \text{ totally ordered chain} \quad S_1 < S_2 < \cdots < S_i < \cdots$$

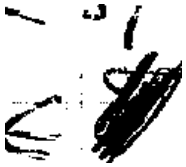
[Crossed out at the end of the line: entry unresolved.]

Uncertain reading. entry unresolved



Original source details: GELFAND-P000988

$S_i \in$ [crossed out: entry unresolved] $\mathcal{F}\}$



Uncertain reading. entry unresolved

Original source details: GELFAND-P000988

[Below $\mathcal{F}_{\max}$:] (Flags)

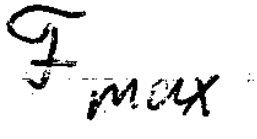


Diagram and typed description. There is no distinct bar above $\mathcal{F}_{\max}$.

Original source details: GELFAND-P000988

trans. with respect to G

(ii)

$$\forall S: \ \dim(S) = \text{length of a max flag}, \quad < +\infty$$

[Below S, $\mathcal{F}$ is added with an upward arrow.]

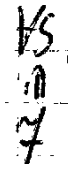


Diagram and typed description. The arrow points up from $\mathcal{F}$ to S.

Original source details: GELFAND-P000988

[Crossed out below $\dim(S)$: entry unresolved.]



Uncertain reading. entry unresolved

Original source details: GELFAND-P000988

[Below "length of a max flag": unresolved entry crossed out, then:] ending in S.



Uncertain reading. unresolved entry

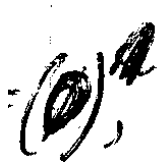
Original source details: GELFAND-P000988

(iii)

$$S, S' \subseteq T, \quad \dim S + \dim S' > \dim(T) \Rightarrow S \cap S' \neq \bigcap_{\mathcal{F} \ni T'} T'$$

(iv) $\forall S \in \mathcal{F}$, if $\{x\}^{\mathrm{cl}} \cap S = (\varnothing)^{\mathrm{cl?}}$, then

$$G^S \cap G^{\{x\}^{\mathrm{cl}}} \underset{\text{maps onto}}{\longrightarrow} \overline{G^S}$$



Uncertain reading. $(\varnothing)^{\mathrm{cl}}$

Original source details: GELFAND-P000988

Then

$$G = \coprod_W G^f w G^f \qquad \text{(disjoint union)}$$

(double coset)

where

$$G^f = \bigcap_I G^{f(i)}, \qquad f : I \to \mathcal{F}$$

[A leader from G^f to the caption:] stabilizer of a max flag.

$$W = N/T$$

[“Here” is written above f.]

$$\{f : I \to \mathcal{F}; \quad S_1 < S_2 < S_3 < S_4 < \cdots$$

— any max chain }

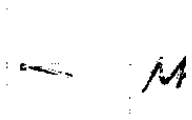


Diagram and typed description. A short stroke precedes “any”. No separate arrowhead is discernible.

Original source details: GELFAND-P000988

GELFAND-P000989

Problems. (I.M.G.)

Describe the 2-sided cosets

1) $U(n)\backslash GL(n)/U(n)$
2) $B(n)\backslash GL(n)/B(n)$
   triang.
3) $U(n)\backslash GL(n)/B(n)$

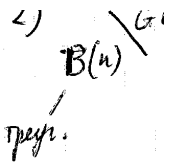


Diagram and typed description. An oblique leader connects “triang.” to the left $B(n)$ in item 2.

Original source details: GELFAND-P000989

Every max flag is obtained as follows:

$$f(i) = \mathrm{cl}(\{e_j\}_{j \le i}), \quad \text{where } \{e_j\} \text{ are points in } X.$$

$$C_i = \mathrm{cl}(\{e_i\})$$

$$N = \{g \mid g \text{ permutes } \{C_i\}\}$$

and we have:

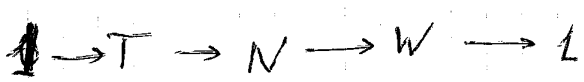


Diagram and typed description. The initial symbol is overwritten. Tentatively 1 over 0.

$[1?] \longrightarrow T \longrightarrow N \longrightarrow W \longrightarrow 1$

Original source details: GELFAND-P000989

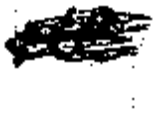
T consists of those that leave each C_i in place;

$W = N/T$ by def.

The speaker knows of no examples other than the classical ones where all the axioms hold.

But: there are examples where the decomp. [crossed out: not (?)] exists, although the axioms do not hold.

Uncertain reading. not (?)



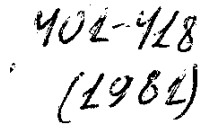
Original source details: GELFAND-P000989

Zelevinsky.

On the paper by S. Abeasis, A. Del Fra, H. Kraft.

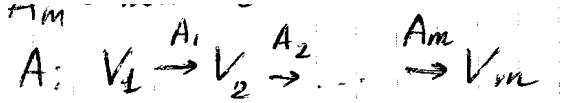
The geometry of repres. of A_m, Math. Ann., 256, N.3, 401–418, (1981).

Diagram and typed description. The opening page reads 401.



Original source details: GELFAND-P000989

A_m — a quiver



Original source details: GELFAND-P000989

Diagram and typed description. The last arrow has a right-pointing head and additional strokes at the left end. Its label is A_m.

$$A: \quad V_1 \xrightarrow{A_1} V_2 \xrightarrow{A_2} \cdots \xrightarrow{A_m} V_m$$

— this is a rep. of a quiver of series A_m.

Here the group acts

$$GL(V_1) \times \cdots \times GL(V_m).$$

Diagram and typed description. The first subscript reads 1.

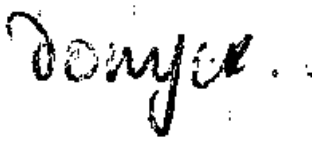


Original source details: GELFAND-P000989

Orbits are reps. up to equiv.

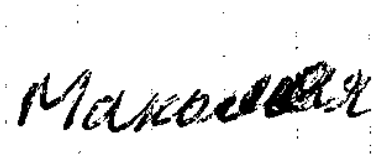
There are finitely many orbits. They admit[?] an explicit descr.

Uncertain reading. admit



Original source details: GELFAND-P000989

Their closures are good varieties (normal, Cohen—Macaulay[?], etc.).



Original source details: GELFAND-P000989

Uncertain reading. Macaulay

GELFAND-P000990

[source number: 16]

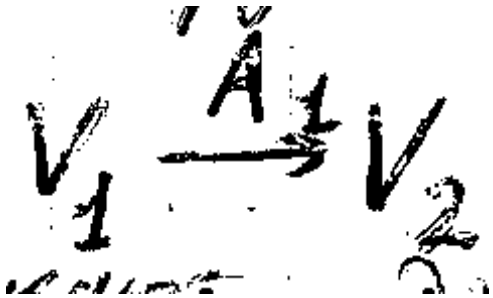
[Crossed out: Their (?)] They are analogous to Schubert cells, and the polynomials describe reps. of p-adic groups, as the geom. of Schubert cells is connected with reps. of classical groups.



Original source details: GELFAND-P000990

Uncertain reading. Their (?)

For



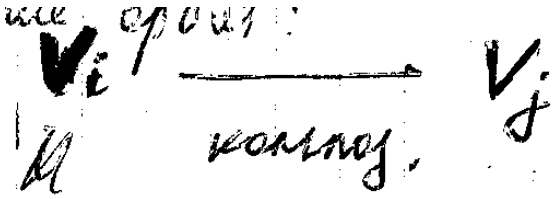
Original source details: GELFAND-P000990

Diagram and typed description. The arrow from V_1 to V_2 is labelled A_1.

$$V_1 \xrightarrow{A_1} V_2$$

determinantal varieties arise.

Description of orbits:



Original source details: GELFAND-P000990

Diagram and typed description. A separate, overtraced mark lies below V_i. Its reading is unresolved.

$$V_i \xrightarrow[\text{compos.}]{} V_j$$

[?]

Specify the ranks of all possible compositions r_{ij}.

A system of inequalities must hold.

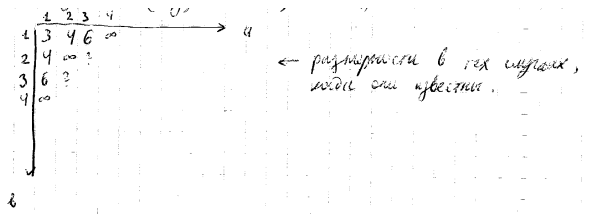
There is another descr. via multiplicities of occurrences.

19/X-81. Gelfand Sem.

Kirillov.

How is the Lie algebra with generators x, y and rels. structured?

$$(\operatorname{ad} x)^{a+1} y = 0, \qquad (\operatorname{ad} y)^{b+1} x = 0, \qquad a, b \in \mathbb{N}.$$



Original source details: GELFAND-P000990

Diagram and typed description. The a axis points right and b down. The entries $(a, b) = (3, 2), (2, 3)$ are question marks. Blank entries remain blank.

b \ a	1	2	3	4
1	3	4	6	∞
2	4	∞	?	$\leftarrow$
3	6	?		
4	∞			

dimensions in the cases where they are known.

Without relations, the number of commutators of bidegree k, ℓ is:

$$d(k, \ell) = \frac{1}{k + \ell} \sum_{d | (k, \ell)} \mu(d) \, \frac{(\frac{k}{d} + \frac{\ell}{d})!}{\frac{k}{d}! \, \frac{\ell}{d}!}.$$

$k = n$, $\ell = n + 1 \Rightarrow$ the nth Catalan number

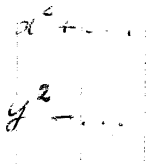
(= the number of ways of placing parentheses in the nonassoc. product $x \cdot x \cdot \ldots \cdot x$.)

[At left above the next line: $(\sim)$.]

Another question: the envelop. algebra.

Rels.

$$0 = x^{a+1} y - \binom{a+1}{1} x^a y x + \binom{a+1}{2} x^{a-1} y x^2 + \ldots,$$
$$0 = y^{b+1} x - \binom{b+1}{1} y^b x y + \binom{b+1}{2} y^{b-1} x y^2 - \ldots.$$



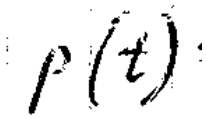
Original source details: GELFAND-P000990

Diagram and typed description. Before the ellipsis the first line has $+$ and the second $-$.

V — a vector space, $S(\bar{V}) = e^{\bar{V}}$ — the symm. algebra

[Below "V — a vector space": "(graded)".]

$p(t) = \sum t^k \dim V_k$ — the Hilbert polynomial.



Uncertain reading. $p(t)$

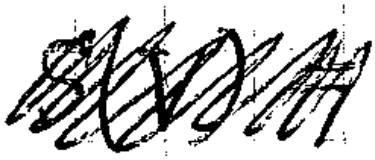
Original source details: GELFAND-P000990

GELFAND-P000991

k — a weight $\Rightarrow$ a weight arises on $S(V) \Rightarrow$ grading on $S(V) \Rightarrow$ the Hilbert polynomial is calculated $P(t)$ for $S(V)$ (or a formal series).

Relation:

[Crossed out: formula unresolved.]



Uncertain reading. formula unresolved

Original source details: GELFAND-P000991

$$\ln P(t) = \sum \frac{C_k}{k} t^k;$$

[Crossed out: short entry unresolved.]



Uncertain reading. short entry unresolved

Original source details: GELFAND-P000991

$$p(t) = \sum \frac{c_k}{k} t^k, \qquad C_k = \sum_{d|k} c_d.$$

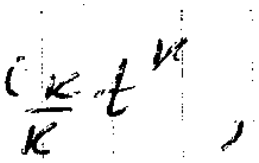


Diagram and typed description. In the expression for $p(t)$ the denominator is k and the numerator is c_k.

Original source details: GELFAND-P000991

$$\underline{x^{k_1}y^{\ell_1}x^{k_2}y^{\ell_2}\dots x^{k_n}y^{\ell_n}}$$

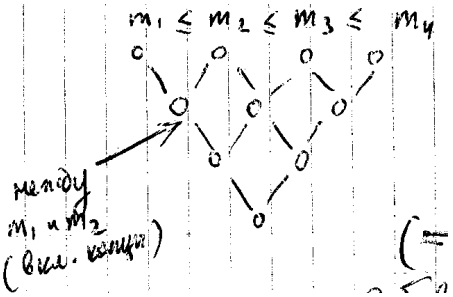
$$k_1+k_2+\dots+k_n=k,$$
$$\ell_1+\ell_2+\dots+\ell_n=\ell.$$

Lexicographic ordering.

y is higher than x. $x<y$.

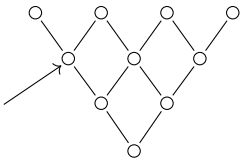
I.M.G. combin. problem

$m_1\le m_2\le m_3\le m_4$ — fixed



Original source details: GELFAND-P000991

Diagram and typed description. Four rows of circles with $4, 3, 2, 1$ entries. The pointer from the note at left marks the first circle of the second row.



between m_1 and m_2 (incl. endpoints)

how many tables?

(= how many reps. of $SL(4)$ with a given highest weight?)

see the appendix to the Gelfand–Minlos–Shapiro book.

GELFAND-P000992

16.11.81. Gelfand Sem.

A.S. Schwarz. Geometry of supergravity.

$\mathbb{R}^{p,q}$ — superspace $(x^1,\dots,x^p,\theta^1,\dots,\theta^q)$.

Λ — Grassmann alg. [Above the abbreviation: r or $4^?$.]



Uncertain reading. r or 4

Original source details: GELFAND-P000992

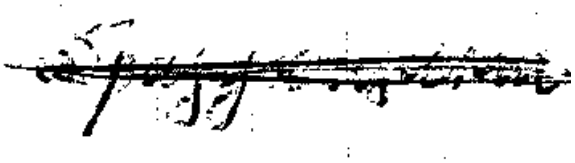
[Crossed out: entry unresolved.]



Uncertain reading. entry unresolved

Original source details: GELFAND-P000992

[Crossed out: with grading$^?$.]



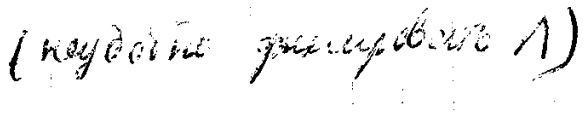
Uncertain reading. with grading

Original source details: GELFAND-P000992

(over $\mathbb{R}$).

$\mathbb{R}^{p,q}_\Lambda$ — the space of rows $(x^1, \dots, x^p, \theta^1, \dots, \theta^q)$, where x^j are even elems. of Λ, θ^j are odd elems. of Λ.

We need to consider this as a functor of Λ (it is inconvenient? to fix Λ).



Uncertain reading. inconvenient

Original source details: GELFAND-P000992

This gives $\mathbb{R}^{p,q}$.

$\mathbb{C}^{p,q}$ is def. analogously and an involution is def. in it.

Consider $\mathbb{C}^{4,2}$ $\quad (x^m, \theta^\alpha) = z^A$.

$$\dim_{\mathbb{C}} \mathbb{C}^{4,2} = (4,2), \qquad \dim_{\mathbb{R}} (\mathbb{C}^{4,2}) = (8,4).$$

In it consider a real subvariety

$$\begin{array}{ll} \Omega & \subset \mathbb{C}^{4,2} \\ (4,4) & \leftarrow \underline{\mathbb{R}}\text{-dim} \end{array}$$

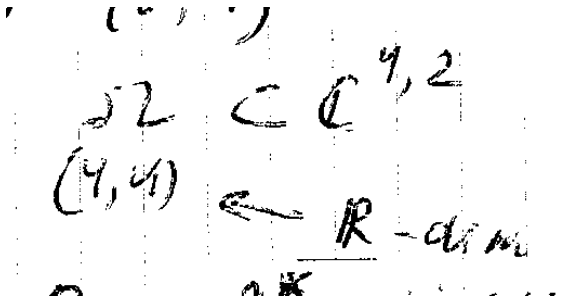


Diagram and typed description. Below Ω the manuscript has $(4,4)$, with an arrow pointing to it from $\underline{\mathbb{R}}$-dim.

Original source details: GELFAND-P000992

Ω is cut out by the eqs.

$$f^k(z, \bar{z}) = 0,$$

f^k are even and real, $\quad k = 1, 2, 3, 4.$

Consider analytic transformations in $\mathbb{C}^{4,2}$:

$$\begin{cases} \widetilde{x}^a = \lambda^a(x, \theta), \\ \widetilde{\theta}^\alpha = \lambda^\alpha(x, \theta) \end{cases} \qquad \text{(indep. of } \bar{x}, \bar{\theta}).$$

GELFAND-P000993

[source number: 40]

They form a supergroup L.

(The group depends on Λ and all together — a functor from Grassmann algs. to ordinary groups — supergroups.)

Consider those that preserve the supervolume.

$$\widetilde{z}^A = \lambda^A(z).$$

Preservation of volume:

$$\operatorname{Ber} \frac{\partial \widetilde{z}^A}{\partial z^B} = 1.$$

We shall consider the functional

$$S(\Omega) = \int A(\text{first, second derivs. of } f^k) \prod_k \delta\big(f^k(z,\bar{z})\big)\, dz\, d\bar{z}.$$

[In the argument of A, before f^k, crossed out: $f^{k?}$.]



Uncertain reading. f^k

Original source details: GELFAND-P000993

[A leader to δ is labelled "δ-fn.".]

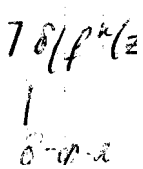


Diagram and typed description. The gloss "δ-fn." is below δ.

Original source details: GELFAND-P000993

Impose a condition on A so that $S(\Omega)$ depends only on Ω, rather than on its eqs.

In classical field theory, in place of the field φ one can consider the graph of the fn. — a surface.

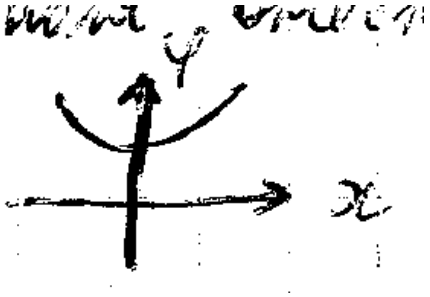
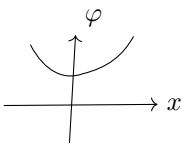


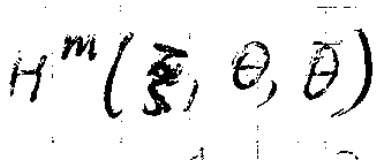
Diagram and typed description. A graph with its minimum above the horizontal x-axis and a vertical φ-axis.



Original source details: GELFAND-P000993

Consider Ω as the graph of the field

$$\mathbb{R}^{4,4} \longrightarrow \mathbb{R}^{4,0}, \qquad H^m(\xi, \theta, \bar{\theta}).$$



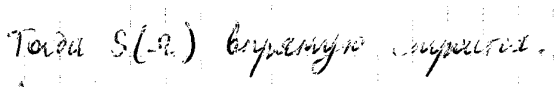
Uncertain reading. The first argument is heavily retraced. The tentative reading is ξ.

Original source details: GELFAND-P000993

The surf. Ω and the superfield are in one-to-one corresp.

Supergravity action: the unique (up to a factor) functional of this kind invariant with respect to the group L.

Then by varying $S(\Omega)$ we obtain[?].



Uncertain reading. obtain

Original source details: GELFAND-P000993

Namely:

$$A = |\det \Gamma|^{-1/2}, \qquad \Gamma \text{ is the Levi form of the surface } \Omega.$$

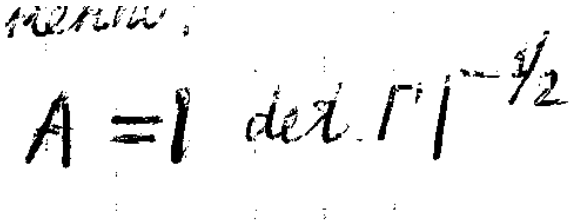


Diagram and typed description. A minus sign precedes the fraction $1/2$.

Original source details: GELFAND-P000993

["(super.)" is inserted beneath "form".]

The Levi form is the first struct. form of the CR-structure.

The form

$$\left.\frac{\partial^2 f^k}{\partial z^A \partial \bar{z}^B}\, dz^A\, d\bar{z}^B \right|_{\substack{\text{on the compl. tangent plane} \\ \frac{\partial f^k}{\partial z^A} dz^A = 0}}$$

(vector-valued)

[Crossed out: ∂f.]

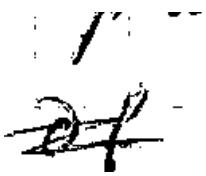


Diagram and typed description. Above "called the Levi form", ∂f is crossed out.

Original source details: GELFAND-P000993

is called the Levi form Γ.

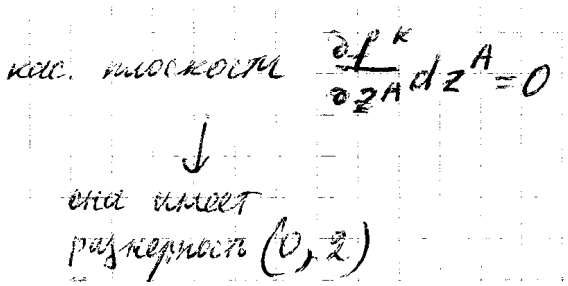


Diagram and typed description. A downward arrow from the words about the tangent plane leads to the dimensional note.

Original source details: GELFAND-P000993

it has dimension $(0, 2)$.

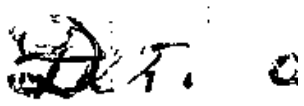
$$\frac{\partial f^k}{\partial x^a}\, dx^a + \frac{\partial f^k}{\partial \theta^\alpha}\, d\theta^\alpha = 0$$

$$\Rightarrow dx^a \text{ is expressed in terms of } d\theta^\alpha \Rightarrow \text{dim. } (0, 2).$$

GELFAND-P000994

The basis in the compl. tangent plane is chosen in different ways.

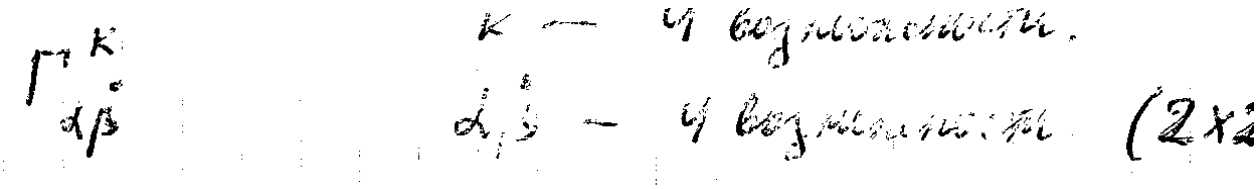
Det. (?) depends on this.



Uncertain reading. The initial strokes of “Det.” are overwritten.

Original source details: GELFAND-P000994

$$\Gamma^k_{\alpha\dot\beta} \qquad \begin{matrix} k \text{ — 4 possibilities,} \\ \alpha, \dot\beta \text{ — 4 possibilities} \quad (2 \times 2 = 4). \end{matrix}$$



Original source details: GELFAND-P000994

Diagram and typed description. In the subscript of Γ the dot is above β.

det — det such a 4×4 matrix.

It and det depend on the choice of basis.

One must [crossed out: illegible] choose the basis correctly in the compl. tangent plane.

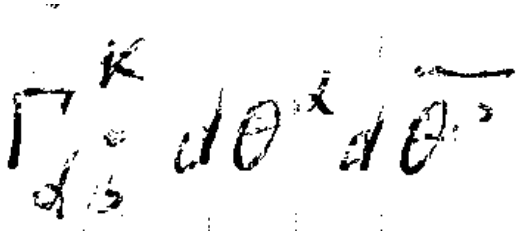


Uncertain reading. The entry after “One must” is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000994

For functions: natural choice of basis

$$\Gamma^k_{\alpha\dot\beta}\, d\theta^\alpha\, d\overline{\theta^\beta}.$$

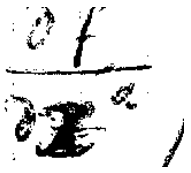


Original source details: GELFAND-P000994

Uncertain reading. The reach of the overbar and the dot at the superscript of the second differential are unclear: $d\overline{\theta^\beta}$ or $d\bar\theta^{\dot\beta}$.

and then one must write

$$A = \left(\det \frac{\partial f^k}{\partial z^a}\right)^{4/3} |\det \Gamma|^{-1/3}.$$



Original source details: GELFAND-P000994

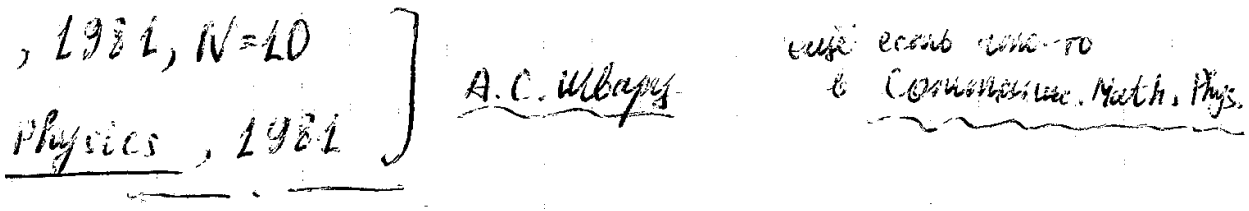
Uncertain reading. The denominator tentatively reads z^a, possibly x^a. The letter is overwritten.

see Yadernaya Fizika, 1981, $N = 10$.

prelim. large (?) work. Nuclear Physics, 1981.

[A right bracket groups the two references, with the label:] A.S. Schwarz.

[At right:] there is also something in Communic. Math. Phys.



Original source details: GELFAND-P000994

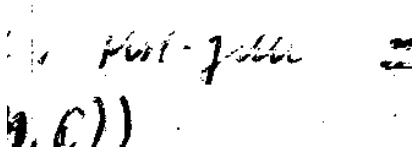
Diagram and typed description. The name and a separate note on a journal stand to the right of the bracket.



Original source details: GELFAND-P000994

Uncertain reading. Before “Nuclear Physics”, tentatively “prelim. large work”.

G-struct. on a compl. manifold (abbrev. uncertain) $\Rightarrow CR$ structure on the surface.



Original source details: GELFAND-P000994

Uncertain reading. After “on a compl.” the manuscript uses a hyphenated abbreviation, tentatively “мн-зии”.

(∼ compl. str. $GL(n, \mathbb{C})$)

Here $SL(4, 2 \mid \mathbb{C})$ is the structure.

L is the autom. group of this struct.

An SCR structure arises on the surf., differing from CR by the det condition.

In $\mathbb{C}^{4,2}$ the coordinates and the field values are mixed up, and they cannot be considered separately.

— · —

GELFAND-P000995

23.11.81 Gelfand Sem.

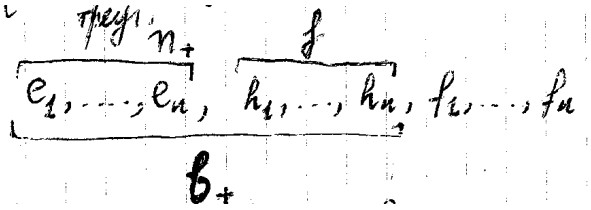
Zelevinsky.

$\mathfrak{g}$ — compl. semis. Lie alg.

$$\mathfrak{g} \supset \underset{\text{Borel}}{\mathfrak{b}} \supset \underset{\text{Cartan}}{\mathfrak{h}} .$$

Example. $\mathfrak{sl}_n$ [below $\mathfrak{b}$: “upper triang.”, below $\mathfrak{h}$: “diag.”].

$$\underbrace{\overbrace{e_1, \dots, e_n}^{\mathfrak{n}_+}, \overbrace{h_1, \dots, h_n}^{\mathfrak{h}}}_{\mathfrak{b}_+}, \quad f_1, \dots, f_n.$$

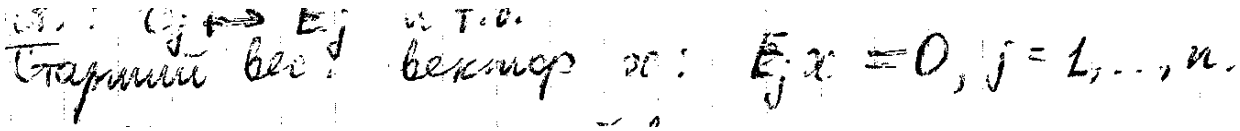


Original source details: GELFAND-P000995

Diagram and typed description. The two upper brackets are marked $\mathfrak{n}_+$ and $\mathfrak{h}$ and the common lower bracket is marked $\mathfrak{b}_+$.

Rep.: $e_j \mapsto E_j$ etc.

Highest weight! vector x: $E_j x = 0$, $j = 1, \dots, n$.



Original source details: GELFAND-P000995

Diagram and typed description. The mark after “weight” reads as an exclamation mark.

$\Rightarrow H_j x$ — also a highest weight.

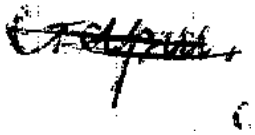
Always $\exists$ in a finite-dimensional rep.

Verma module

$\chi \in \mathfrak{h}^*$.

[Crossed out: "Highest" (abbrev., tentative)] Vector of highest weight χ:

$$\begin{cases} ev = 0, & \forall e \in \mathfrak{n}_+, \\ hv = \chi(h)v, & \forall h \in \mathfrak{h}. \end{cases}$$



Original source details: GELFAND-P000995

Uncertain reading. The crossed-out entry tentatively reads "Highest" in abbreviation.

Verma module: the largest module $M(\chi)$ gen. by highest weight χ.

(Universal module with these props.)

$M(\chi)$ is not irreduc. (generally speaking).

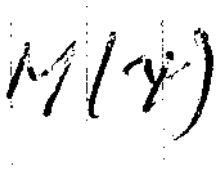
$L(\chi)$ — the unique irreduc. quotient module of $M(\chi)$.

Thm. $\exists$ a finite compos. series

$$M(\chi) = M_1 \supset M_2 \supset \ldots \supset M_n = 0, \qquad M_i/M_{i+1} = L(\chi_i).$$

$\Rightarrow$ every subquotient has this form.

Question: given ψ, what is the multiplicity of occurrence of $M(\psi)$ in the compos. series of a Verma mod.



Original source details: GELFAND-P000995

Diagram and typed description. The question has $M(\psi)$.

(This is equiv. to describing the Verma mod. [inserted above: "itself"].) [Below the parentheses: "— can be calculated".]



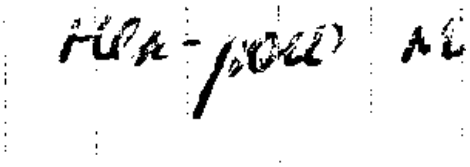
Original source details: GELFAND-P000995

Diagram and typed description. The word "itself" is above the line with an insertion mark.

Denote the multiplicities

$$m(\chi, \psi) = [M(\chi) : L(\psi)].$$

Multiplicities = value of a certain polynomial at pt. 1.



Uncertain reading. The word “certain” before “polynomial” is abbreviated.

Original source details: GELFAND-P000995

Condition:

$$\chi + \rho \in W(\psi + \rho), \quad \text{i.e.} \quad \chi + \rho = w(\psi + \rho).$$

[Label at W: “Weyl gr.”]

ρ is def. by the formula $\rho(h_i) = 1$.

Only then can the multiplicity $m(\chi, \psi)$ be $\neq 0$.

χ integral dominant if $\lambda(h_i) \in \{0, 1, 2, \ldots\}$ for all h_i.

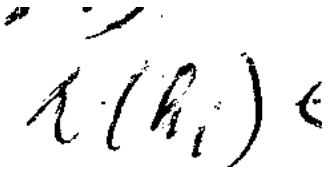


Diagram and typed description. After “if” the condition uses $\lambda(h_i)$, while the line begins with χ.

Original source details: GELFAND-P000995

GELFAND-P000996

[source number: 46]

Our case:

χ_0 — integral dominant, and

$$\chi + \rho = w_1(\chi_0 + \rho)$$
$$\psi + \rho = w_2(\chi_0 + \rho)$$

Then

$$m(\chi, \psi) = m(w_1, w_2)$$

(does not depend on the choice of dominant weight χ_0).

Kazhdan–Lusztig — hypothet. expression.

Now proved: Beilinson, Bernstein. Brylinski, Kashiwara.

They wrote down:

$$P_{w_1,w_2}(q)$$

— a polynomial (with nonneg. coefficients).

Conjecture

$$m(w_1, w_2) = P_{w_1,w_2}(1).$$

$$P_{w_1,w_2}(q) = \sum a_i q^i.$$

Meaning of a_i — ?

Explanation: $\exists$ a Jantzen filtration

$$M(\chi) \supset M_1 \supset M_2 \supset \cdots$$

$a_i =$ multiplicity of $L(\psi)$ in M_i/M_{i+1}.

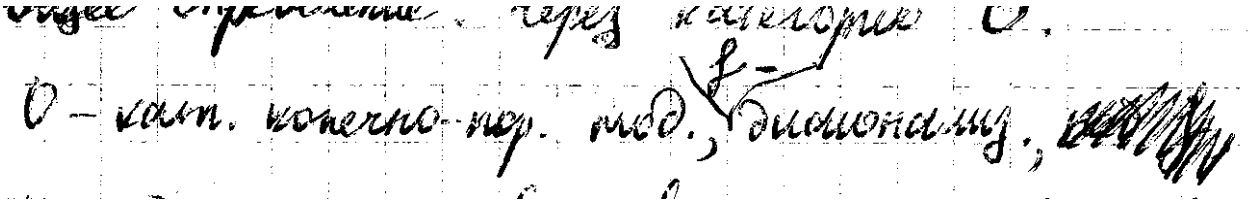
Beilinson.

Jantzen filtration.

Dual module ${}^*M(\chi)$.

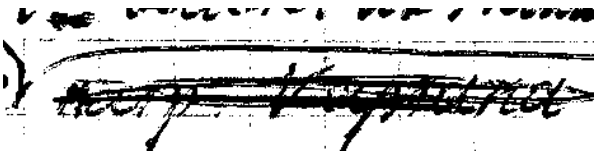
General definition: through category O.

O is the cat. of finitely gen. mod., [added above the line: $\mathfrak{h}$-]diagonaliz., [crossed out: illegible], where [crossed out: illegible] [replacement above the line: $\mathfrak{n}_-$ (?) acts locally nilpotently].



Original source details: GELFAND-P000996

Uncertain reading. Dense strokes cross out the entry after “diagonaliz.”.



Original source details: GELFAND-P000996

Uncertain reading. The entry after “where” is crossed out. Its exact reading is unresolved.

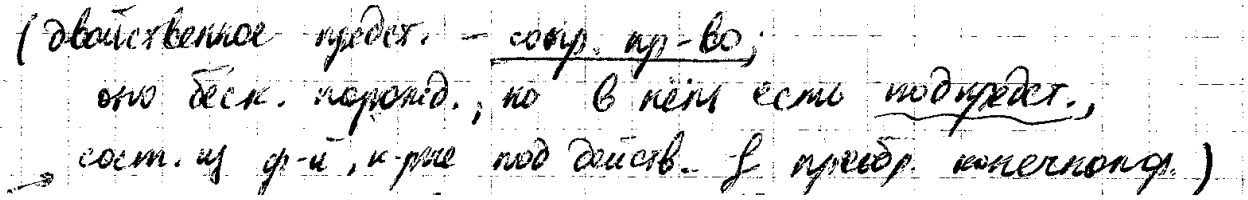


Original source details: GELFAND-P000996

Uncertain reading. The sign after $\mathfrak{n}$ is overwritten. The reading “$_-$” is tentative.

It has a natural involution $* : O \longrightarrow O$.

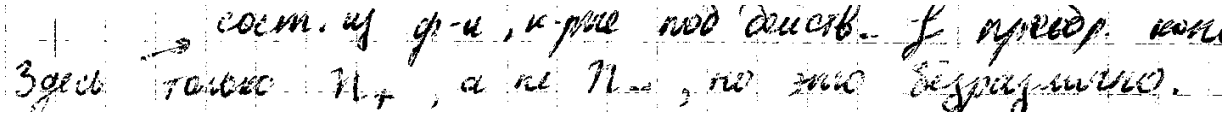
(The dual rep. is the dual space. It is infinitely gen., but contains a subrep. consisting of functions that transform finite-dimensionally under $\mathfrak{h}$.)



Original source details: GELFAND-P000996

Diagram and typed description. The end of the line reads “transform finite-dimensionally”, abbreviated in the manuscript.

[Added with an arrow to the preceding line:] Here only $\mathfrak{n}_+$, and not $\mathfrak{n}_-$, but this makes no difference.



Original source details: GELFAND-P000996

Diagram and typed description. The addition says $\mathfrak{n}_+$ followed by “and not $\mathfrak{n}_-$”.

S. Gelfand.

On each $M(\chi)$ introduce a bilinear form with numerical values.

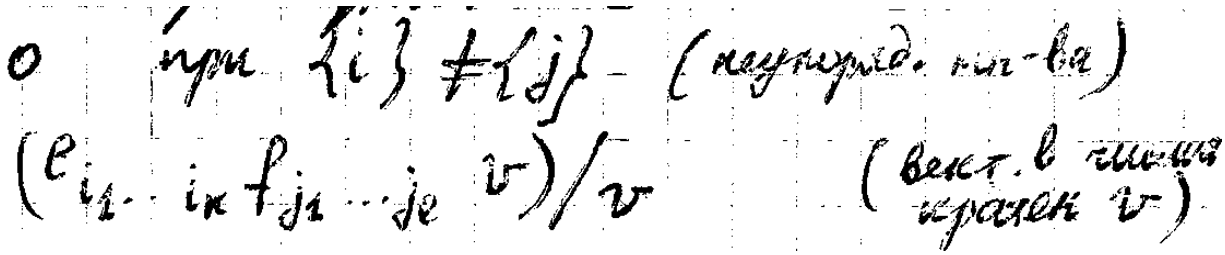
$$M(\chi) \ni f_{i_1} \dots f_{i_k} v, \qquad v \text{ — highest-weight vector.}$$



Original source details: GELFAND-P000996

Diagram and typed description. The weight in the notation before $\ni$ is in parentheses.

$$(f_{i_1} \dots f_{i_k} v,\ f_{j_1} \dots f_{j_l} v) = \begin{cases} 0 \quad \text{when } \{i\} \neq \{j\} & \text{— (unordered sets),} \\ (e_{i_1 \dots i_k} f_{j_1 \dots j_l} v)/v & \text{(vector in numerator is a multiple of } v\text{).} \end{cases}$$



Original source details: GELFAND-P000996

Diagram and typed description. In the numerator each of the single letters e and f carries a sequence of indices.

GELFAND-P000997

We obtain a form invar. with respect to the Lie alg.

[Crossed out: illegible] We will identify all $M(\chi)$ (for different χ) as follows

$$f_{i_1} \dots f_{i_l} v_\chi \longrightarrow f_{i_1} \dots f_{i_l} v_{\chi'}.$$



Uncertain reading. A short entry before “We will” is crossed out.

Original source details: GELFAND-P000997

Consider $M(\chi)$ with [crossed out: illegible] fixed χ and then consider $M(\chi + t\rho)$.



Uncertain reading. A short entry is crossed out between “with” and “fixed”.

Original source details: GELFAND-P000997

We obtain many bilinear forms on one space:

$$(\ ,\ )_t.$$

Regard them all as forms on $M(\chi)$.

For small $t \neq 0$ the form is nondegenerate.

Def.

$$M_i = \{x \in M_\chi \mid (x, y)_t = O(t^i)\}.$$

[Above the condition, with an insertion mark:] $\forall y \in M(\chi)$:

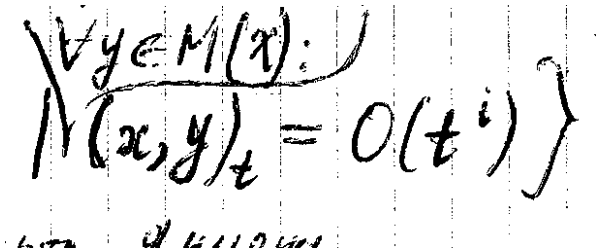


Diagram and typed description. The condition has the insertion $\forall y \in M(\chi)$ followed by a colon.

Original source details: GELFAND-P000997

This is the Jantzen filtr.

It is a filtr. by $\mathfrak{g}$-submodules.

What if we change ρ? Apparently the mult. will remain the same.

Polynomial

$$P_{w_1,w_2}(q) = \sum q^i \big([L_\psi : M_i/M_{i+1}]\big),$$

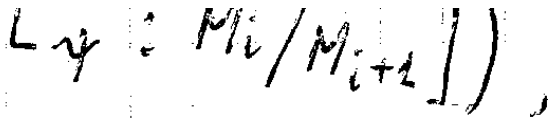


Diagram and typed description. The manuscript puts L_ψ before the colon and M_i/M_{i+1} after it.

Original source details: GELFAND-P000997

where

$$\psi = w_2(\chi_0 + \rho) - \rho,$$
$$M_i \subset M_\chi, \qquad \chi = w_1(\chi_0 + \rho) - \rho.$$

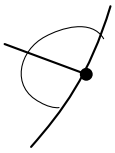
P_{w_1,w_2} — are determined recurr..

$$a_i = \dim H^i(\text{of one Schubert cell near another, according to MacPherson}).$$

(local cohomol. of a normal section)

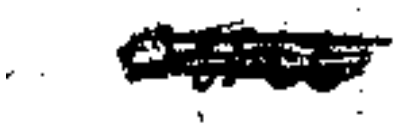


Diagram and typed description. A slanted line with a marked point, a curve and a short segment leading to the point. There are no labels.



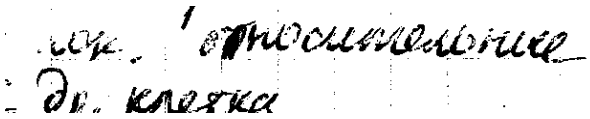
Original source details: GELFAND-P000997

or [crossed out: illegible] local rel. (?)
cell $\subset$ other cell.



Uncertain reading. The word after “or” is crossed out.

Original source details: GELFAND-P000997



Uncertain reading. Tentative reading “local relative”, with the first word abbreviated.

Original source details: GELFAND-P000997

Zelevinsky. Other problems, the same polynoms.



Original source details: GELFAND-P000997

Diagram and typed description. In the heading “the same” is followed by the abbreviation for “polynomials”.

Consider a finite-dimensional irreduc. rep.

$$L(\chi) = \bigoplus_\lambda L(\chi)_\lambda.$$

[Below $L(\chi)_\lambda$, joined by a stroke:] weight space with weight χ.

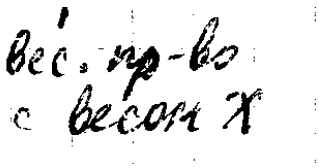
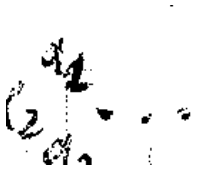


Diagram and typed description. The label ends with the letter χ.

Original source details: GELFAND-P000997

I. M. G. A good fn. for dim $L(\chi)_\lambda$ is known

$$\begin{vmatrix} x_1^{a_1} & x_2^{a_1} & \dots & x_n^{a_1} \\ x_1^{a_2} & x_2^{a_2} & \dots & x_n^{a_2} \\ x_1^{a_n} & x_2^{a_n} & \dots & x_n^{a_n} \end{vmatrix} : \begin{vmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \\ x_1^{n-1} & \dots & x_n^{n-1} \end{vmatrix} = \sum m(\chi,\lambda) x^\lambda.$$



Original source details: GELFAND-P000997

Uncertain reading. The first exponent of x_2 reads as a_1 through the overlapping strokes.

GELFAND-P000998

[source number: 47]

Here

$$\chi = (a_1, \dots, a_n), \quad \lambda = (\lambda_1, \dots, \lambda_n), \qquad x^\lambda = x_1^{\lambda_1} \dots x_n^{\lambda_n}.$$

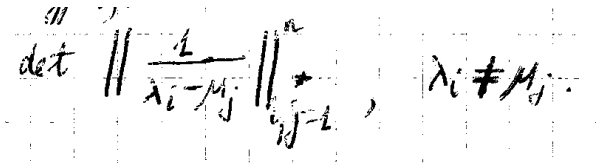
$$m(\chi, \lambda) = \dim L(\chi)_\lambda.$$

Problem (for first-year students).

Calculate

$$\det \left\| \frac{1}{\lambda_i - \mu_j} \right\|_{i,j=1}^{n}, \qquad \lambda_i \neq \mu_j.$$

[A star to the right of the matrix.]



Original source details: GELFAND-P000998

Diagram and typed description. The denominator is $\lambda_i - \mu_j$. A star is written to the right of the double bar.

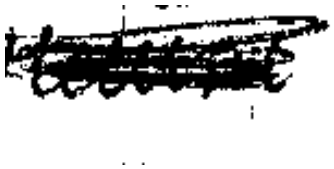
— · —

Zelevinsky.

[Crossed out before the formula: illegible.]

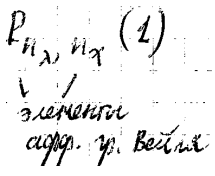
$$m(\lambda, \chi) = P_{n_\lambda, n_\chi}(1).$$

[Strokes from both indices lead to the label:] elements of the aff. Weyl gr.



Original source details: GELFAND-P000998

Uncertain reading. The word before the formula is crossed out with dense strokes.



Original source details: GELFAND-P000998

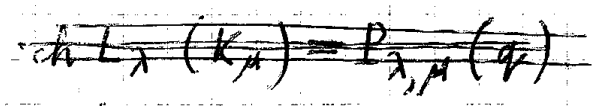
Uncertain reading. The indices of P tentatively read n_λ, n_χ.

Conjecture.

The same ∀ Kac–Moody algebras.

[The entire formula is crossed out.]

$$\operatorname{ch} L_\lambda(K_\mu) = P_{\lambda,\mu}(q).$$



Original source details: GELFAND-P000998

Uncertain reading. The argument reads K_μ through the cancellation.

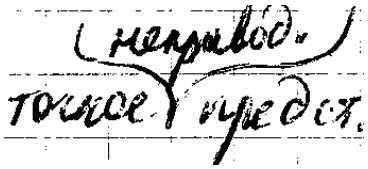
Elashvili.

G — a lin. alg. group, i.e. $G \subset GL(n, \mathbb{C})$.

$L(G)$ — its Lie algebra.

$U(L)$ — the universal enveloping alg.

$U(L)$ is called primitive if the zero ideal is primitive (= the kernel of an irreduc. rep.), i.e. there is a faithful [added above the line: (irreduc.)] rep. of $U(L)$.



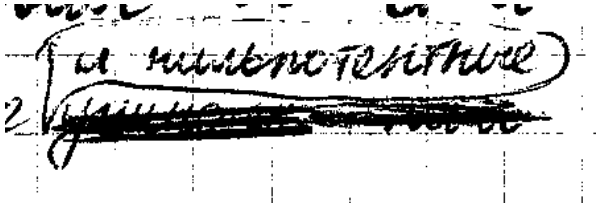
Original source details: GELFAND-P000998

Diagram and typed description. "(irreduc.)" is inserted above "faithful rep." with a bracket.

What conditions must G and $L(G)$ satisfy? (question of Jacobson)

[Below the name:] last year.

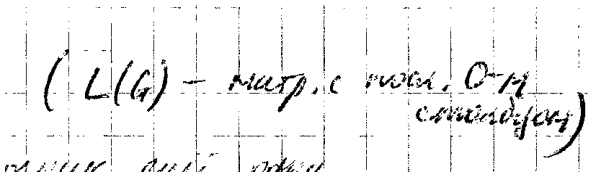
Semisimple ones all [crossed out: unipotent (?)] [added above the line: (and nilpotent ones)] are unsuitable.



Original source details: GELFAND-P000998

Uncertain reading. The crossed-out word tentatively reads "unipotent". Above it, a bracket contains "and nilpotent ones".

Example: 1) Lie alg. of the group of affine transf. ($L(G)$ consists of matr. with last column zero).



Original source details: GELFAND-P000998

Diagram and typed description. The manuscript specifies the last column as zero.

2) $L = gl(2,\mathbb{C}) + V_4$.

[Below V_4, joined by a stroke:] commut. ideal.

$gl(2,\mathbb{C})$ acts on V_4 by the unique 4-dim. irreduc. rep.

[At right after two vertical strokes:] Another interesting Γ-fn. arises here.

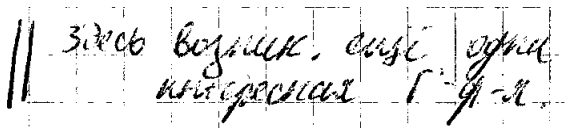
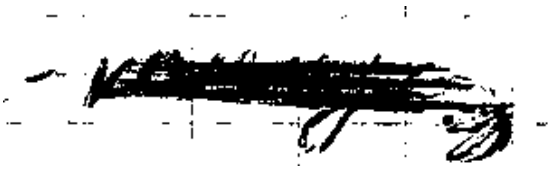


Diagram and typed description. The right-hand note reads “Γ-fn.”.

Original source details: GELFAND-P000998

Thm. There are no others if $L = R + V$, R is [crossed out: illegible] simple, V is commut., and the action of R on V is irreduc.



Uncertain reading. The word after R and before “simple” is crossed out with dense strokes.

Original source details: GELFAND-P000998

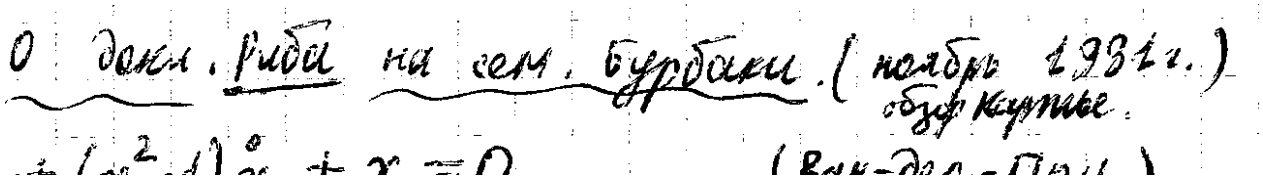
[Below the sum:] semidirect sum.

There is a method of construction beyond this.

GELFAND-P000999

30/XI-81 Gelfand Sem.

I. M. G. On Reeb’s talk at the Bourbaki sem. (November 1981) [Below the date:] Cartier’s survey.



Original source details: GELFAND-P000999

Diagram and typed description. The talk line names Reeb. Below the date is “Cartier’s survey”.

$$\varepsilon\ddot{x} + (x^2 - 1)\dot{x} + x = 0. \tag{1}$$

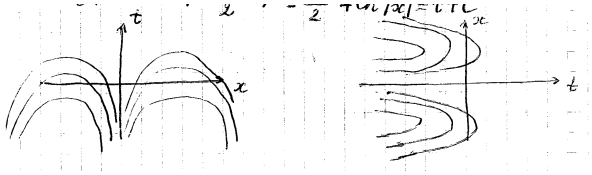
(Van der Pol)

What happens as $\varepsilon \to 0$?

$\varepsilon = 0$:

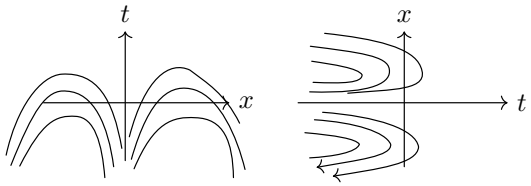
$$\frac{dx}{dt} = -\frac{x}{x^2-1}, \qquad \frac{dx(1-x^2)}{x} = dt, \qquad \frac{dx}{x} - x\,dx = dt.$$

$$C + t = \ln|x| - \frac{x^2}{2}, \qquad -\frac{x^2}{2} + \ln|x| = t + C.$$



Original source details: GELFAND-P000999

Diagram and typed description. The left plot has axes t and x. The right plot has axes x and t. Each plot has three branches on either side of the axis. Two lower branches at right have strokes resembling leftward arrows.

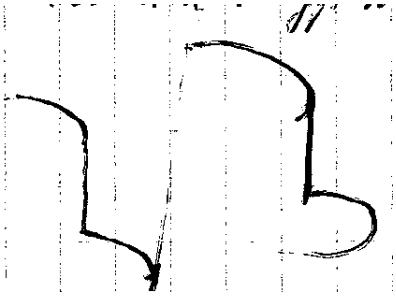


Another remarkable eq.:

$$\varepsilon\ddot{u} + \left(\frac{\dot{u}^3}{3} - \dot{u}\right) + u = 0. \tag{2}$$

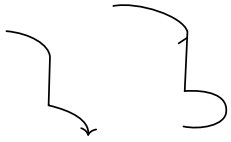
(Rayleigh)

It reduces to the Van der Pol eq.

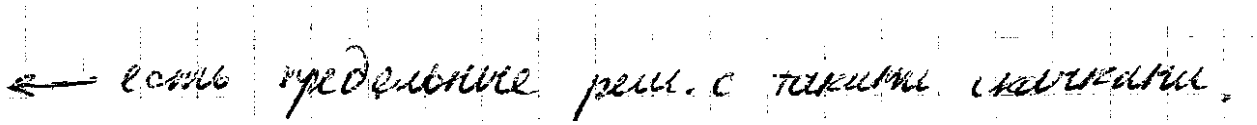


Original source details: GELFAND-P000999

Diagram and typed description. Two curves with vertical segments. At the lower left endpoint is a downward arrow. The short stroke near the upper bend at right is unclear.



[At right, with an arrow pointing left:] there are limiting solns. with such jumps.



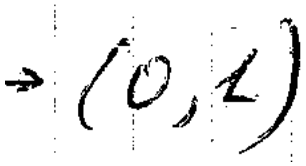
Original source details: GELFAND-P000999

Diagram and typed description. The arrow from the label points toward the two curves.

Eq. (1) is studied using nonstandard analysis!

Ω — a set,

$$F = 2^{\Omega}, \qquad p : F \to (0,1).$$



Original source details: GELFAND-P000999

Diagram and typed description. The interval is written with round brackets: $(0,1)$.

$$p(A \cup B) = p(A) + p(B) \quad \text{when } A \cap B = \varnothing.$$

X — a set,

$$x' : \Omega \to X, \qquad x'' : \Omega \to X.$$

A random element of X is a class of equivalent maps $\Omega \to X$.

[In small writing below the line:]

$$x' \sim x'' \Leftrightarrow P(x' = x'') = 1.$$

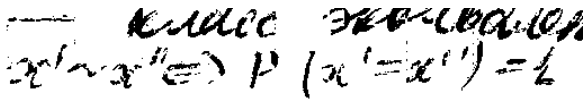


Diagram and typed description. The equivalence condition uses a capital P.

Original source details: GELFAND-P000999

[At right:] x is the equiv. class of x'. $\Rightarrow x'$ is a version of x.

GELFAND-P001000

Kruskal's numbers: the simplest example of a field extending $\mathbb{R}$.

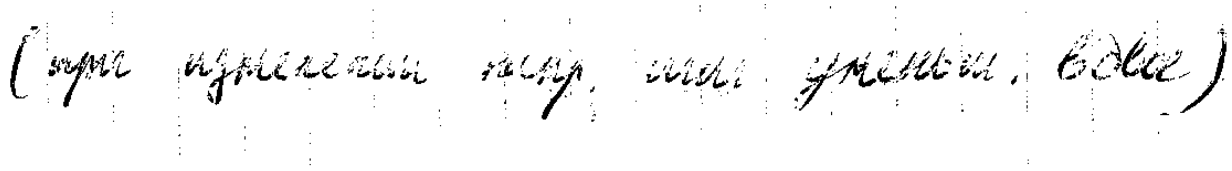


Diagram and typed description. The heading reads "Kruskal's".

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$$\begin{aligned} & \varnothing = 0. \\ \text{up} \quad & \uparrow = 1, \\ \text{down} \quad & \downarrow = -1, \\ & \overbrace{\uparrow\uparrow\uparrow \ldots \uparrow}^{n} = n, \\ & \uparrow\downarrow = \frac{1}{2}, \\ & \uparrow\downarrow\uparrow = \frac{3}{4}. \end{aligned}$$

(when the direction changes the step is halved)



Original source details: GELFAND-P001000

Diagram and typed description. The manuscript says the step is halved when the direction changes.

$\hat{\uparrow}$ — infinite repetition of $\uparrow$.

$$\hat{\uparrow}\downarrow = \infty - 1; \qquad \hat{\uparrow} = \infty.$$

[A sign before $\infty - 1$ is crossed out.]



Uncertain reading. Strokes to the left of ∞ obscure the crossed-out sign.

Original source details: GELFAND-P001000

(there is no predecessor)

$$\hat{\uparrow}\hat{\downarrow} = \frac{1}{2}\infty.$$

$\hat{\uparrow} : \hat{\uparrow}\hat{\downarrow} = 2$ — exact thm.

$$\hat{\hat{\uparrow}} = \infty^2,$$

$$\uparrow\hat{\downarrow} = \varepsilon.$$

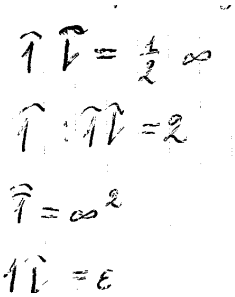
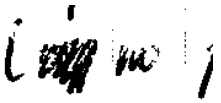


Diagram and typed description. In the last pair of formulas the first arrow has two arcs and the downward arrow has one.

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Krichever.

([Crossed out: illegible] from (?) work by Brozovskii, Dzyaloshinskii and Krichever)



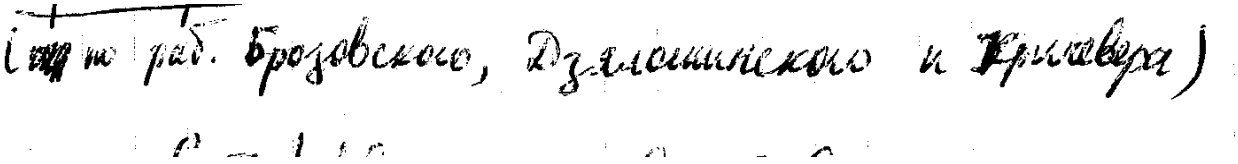
Uncertain reading. A short entry is crossed out at the start of the parenthesis.

Original source details: GELFAND-P001000



Uncertain reading. The preposition between the cancellation and “work” tentatively reads «по».

Original source details: GELFAND-P001000



Original source details: GELFAND-P001000

Diagram and typed description. The surname is written «Брозовского», with «о» in the first syllable.

$$G = \bigcup_n G_n, \qquad G_n \subset G_{n+1}.$$



Uncertain reading. A short entry follows G_{n+1}. Its reading is unresolved.

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$$u(z), \qquad \Gamma \ni z.$$

[Below $u(z)$, joined by a stroke:] matrix fn.

[Below Γ, joined by a stroke:] torus ($\mathbb{C}$/lattice).

$$u^{ij}(z) = \frac{\sigma(z - a_1^{ij}) \dots \sigma(z - a_n^{ij})}{\sigma^n(z)}\, e^{\zeta(z)(x_i - x_j)},$$
$$a_1^{ij} + \dots + a_n^{ij} = x_i - x_j.$$

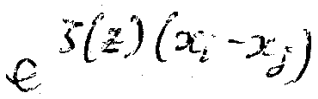


Diagram and typed description. The exponent reads $\zeta(z)(x_i - x_j)$.

Original source details: GELFAND-P001000